PRANAV DUBEY

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SUMMARY

Final-year CS student specializing in computer vision and generative AI. Co-authored a YOLOv8 flood-detection model with IIT Delhi (**93.0% mAP50**, under journal review) and built a CCTV anomaly-detection system beating a published baseline (**0.803** vs. **0.754** AUC-ROC). Proficient in Python, PyTorch, RAG/LLM pipelines, and FastAPI backend deployment; seeking an applied ML/AI engineering role.

EDUCATION

Vellore Institute of Technology, Bhopal **2023 – 2027**
B.Tech in Computer Science Engineering *CGPA: 8.07*

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, FastAPI, Streamlit.
- **Generative AI & LLMs:** LLM Integration, Context Engineering, RAG, LangChain, ChromaDB, Ollama, Hugging Face.
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, CNNs, LSTMs, YOLOv8, NLP.
- **Data & Analytics:** NumPy, Pandas, Matplotlib, SQL, PostgreSQL.
- **MLOps & Tools:** AWS, Docker, Git, REST APIs, Postman.

RESEARCH EXPERIENCE

AQUANET: YOLOv8-based Detection of Objects Displaced During Floods *Co-author | Under Journal Review*

- Co-authored a **YOLOv8**-based computer vision model with **Prof. Brejesh Lall** (IIT Delhi) and PhD scholar Gurjant Singh for detecting high-value objects submerged in post-flood water and mudflow, achieving **93.0% mAP50**.
- Benchmarked **YOLOv8 s/m/l** variants by reproducing training runs and validating accuracy, precision, recall, and mAP, confirming reproducibility of the reported results.
- Prepared the training dataset, ran ablation studies, and co-wrote the manuscript now **under journal review**, collaborating across the full research lifecycle from data to submission.

PROJECTS

Congenital Heart Defect Prediction System | *Java, Spring Boot, FastAPI, PostgreSQL, PyTorch, Docker* **Demo** | **GitHub**

- Trained a dual-branch **ConvNeXt** for chest X-ray classification (Normal, ASD, VSD) using **CLAHE**, feature fusion, and focal loss, achieving **82.6% accuracy** and **0.83 macro F1**.
- Designed and shipped a **FastAPI + Spring Boot** backend exposing **22+ REST APIs** on **PostgreSQL** and Docker, serving predictions and patient history end-to-end.
- Secured medical data access end-to-end with **JWT authentication** and **role-based access control (RBAC)**, restricting actions by clinician/admin role.

Cross-Source News Verification | *Python, LangChain, ChromaDB, Ollama, FastAPI, React* **GitHub**

- Designed a **RAG** pipeline that retrieves related news coverage and flags where independent sources agree, conflict, or offer no supporting evidence, with grounded citations for every claim.
- Improved retrieval precision by removing duplicate syndicated articles and tuning similarity thresholds to suppress low-relevance matches.
- Implemented local inference with **Ollama** and **ChromaDB**, built modular **LCEL** components, wrote **225** offline tests, and shipped the app via **FastAPI** and React.

AI Video Surveillance: Anomaly Detection System | *PyTorch, YOLOv8, OpenCV, FastAPI, Docker* **Demo** | **GitHub**

- Engineered an end-to-end CCTV anomaly detection system (**ResNet50 + LSTM**) trained on **1.37M+ frames** from **1,900 UCF-Crime videos**.
- Outperformed the **Sultani et al. (2018)** baseline by **6.5 points**, reaching **0.803 AUC-ROC** (vs. **0.754**), with **YOLOv8** and **Grad-CAM** for visual explainability.
- Deployed the model behind a **FastAPI** backend supporting asynchronous video upload and inference, live on a Hugging Face Spaces demo.

ACHIEVEMENTS

- Scored **97%** in the NPTEL Marketing Analytics certification conducted by IIT Kharagpur, ranking among the top **5%** of certified learners.
- Participated in the **Shell.ai Hackathon 2025**, developing and optimizing machine learning models for sustainable aviation fuel blend property prediction in a national-level industry challenge.